Information for Candidates

Electronic charge: $e = 1.60 \times 10^{-19}$ C

Boltzmann constant: $k = 1.38 \times 10^{-23}$ J/K

Permittivity of free space: $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

Permeability of free space: $\mu_0 = 4\pi \times 10^{-7}$ H/m

Switch-Mode Power Supplies

Voltage ratio equations for buck SMPS $\frac{V_O}{V_I} = \delta$ or $\frac{V_O}{V_I} = \frac{1}{1+\frac{2fL I_O}{V_I \delta^2}}$

Voltage ratio equations for boost SMPS $\frac{V_O}{V_I} = \frac{1}{1-\delta}$ or $\frac{V_O}{V_I} = \frac{1}{1-\frac{V_I \delta^2}{2fL I_I}}$

Three-Phase Systems

Line Voltages and Current

Star $V_L = \sqrt{3}\, V_P;\ I_L = I_P$ Delta $V_L = V_P;\ I_L = \sqrt{3}\, I_P$

Power $P_{3\phi} = 3\, V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Induction Machines

$\omega_S = {}^{\omega_E}/_{P}$; $s = \frac{\omega_S - \omega_R}{\omega_S}$; $T_{EM} = \frac{3\, I'^2_R\, R'_R}{\omega_R}\left(\frac{1-s}{s}\right)$

Power Flow in Lines and Cables

Cable Parameters (expressed per metre)

$R'_{LF} = \frac{1}{\sigma_C \pi r_C^2} + \frac{1}{\sigma_C 2\pi r_O\, t_O}$ $L' = \frac{\mu}{2\pi} ln\left({}^{r_O}/_{r_C}\right)$ $C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({}^{r_O}/_{r_C}\right)}$ $G' = \frac{2\pi\sigma_I}{ln\left({}^{r_O}/_{r_C}\right)}$

OHL Parameters in approximate form (expressed per metre)

$R'_{LF} = \frac{2}{\sigma_C \pi r_C^2}$ $L' = \frac{\mu}{\pi} ln\left({}^{d}/_{r_C}\right)$ $C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({}^{d}/_{r_C}\right)}$ $G' = \frac{\pi\sigma_I}{ln\left({}^{d}/_{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{{V_S}^2}{Z_{SR}} cos(\theta) - \frac{V_R V_S}{Z_{SR}} cos(\theta + \delta) \qquad Q_S = \frac{{V_S}^2}{Z_{SR}} sin(\theta) - \frac{V_R V_S}{Z_{SR}} sin(\theta + \delta)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\, P_S + X\, Q_S}{|V_S|}$

Photovoltaic Systems

$$I_{PV} = I_{Ph} - I_{AK} - I_{Sh} \qquad I_{AK} = I_0\left(e^{\frac{V_{PV} + I_{PV} R_S}{K_I\, v_T}} - 1\right)$$

Answers

1. This question covers several topics and all parts should be attempted.

a) An electricity grid is divided into a transmission grid and a distribution grid.

i) Explain their different roles. [4]

[Interpretation of note and formulation of argument.]

[2 marks for accurate expression of each of the two points.]

- Long distance conveyance of bulk energy is role of transmission, essentially locational arbitrage of energy, and providing access to diverse sources of energy.
- Onward conveyance of energy to large numbers of consumers through local connections from a bulk supply point is the role of distribution. This is changed somewhat by rise in export of power from distributed generation.

ii) Explain why a transmission system is constructed with meshed double circuits and a distribution system with radial single circuits. [4]

[Interpretation of note and formulation of argument.]

[2 marks for accurate expression of each of the two points.]

- Transmission outages have an impact on a large number of customers and the cost of such outages justifies security through asset redundancy such as duplication of circuits and alternative paths.
- Low number of customers affected by a circuit outage in distribution only justifies restoration of supply through switching or repair.

b) Consider an overhead line of 3 km used for a single-phase AC supply of 11 kV at 50 Hz and with two conductors of aluminium (conductivity of 35×10^6 S/m) each with a radius of 5.0 mm. The conductors are spaced 1.0 m apart on the cross-arms of the towers.

i) Explain why one expects an overhead line to have a low capacitance per unit length and a high inductance per unit length compared to a cable of equivalent voltage and current rating. [4]

[Interpretation and formulation of an argument.]

OHL are air-insulated with a large separation distance (about a metre) giving them a low capacitance compared with a cable which is plastic insulated (relative permittivity of 3 – 4) and separation distance of 1 - 10 cm. Capacitance is therefore 30 – 400 times lower in an OHL. [2 marks]

The loop area of an OHL is larger than that of a cable because of the great separation distance and therefore the flux-linkage (with its own flux) is greater and the inductance greater. [2 marks]

ii) Calculate the inductive reactance and resistance of the route (using the approximate form of the equations). [4]

[Use of standard formula.]

For inductance.

$$L' = \frac{\mu}{\pi} ln\left({}^{d}/_{r_c}\right) = \frac{4\pi \times 10^7}{\pi} ln\left(\frac{1.0}{0.005}\right) = 2.12 \times 10^{-6}\ H/m$$

Answers

$$L = l\,L' = 3{,}000 \times 2.12 \times 10^{-6} = 6.4\,mH$$

$$X = 2\pi f\,L = 2.00\,\Omega$$

[2 marks]

For resistance.

$$R'_{LF} = \frac{2}{\sigma_C \pi\, r_C^2} = \frac{2}{35 \times 10^6 \times \pi \times 0.005^2} = 7.28 \times 10^{-4}$$

$$R = l\,R' = 3{,}000 \times 7.28 \times 10^{-4} = 2.18\,\Omega$$

[2 marks]

iii) Estimate the voltage magnitude difference between sending and receiving ends of the overhead line when real power of 1 MW is consumed at the receiving end with a power factor of 0.95 lagging. [4]

[Applying standard approximation together with using understanding of relation between real power, power factor and reactive power.]

$$\theta = \cos^{-1}(PF) = \pm 23.1°$$

Positive value of angle is taken for lagging power factor.

$$S = \frac{P}{PF} = \frac{1 \times 10^6}{0.95} = 1.087\,MVA$$

$$Q = S\sin(\theta) = 1.087 \times 10^6 \sin(23.1°) = 0.426\,MVAr$$

[2 marks for correct method and value of reactive power]

$$\Delta V \approx \frac{R\,P + X\,Q}{V} = \frac{2.18 \times 1 \times 10^6 + 2.00 \times 0.426 \times 10^6}{11{,}000} = 276\,V$$

[2 marks for accurate value of voltage difference.]

c) Consider a three-phase AC supply with a line voltage of 400 V.

i) Three impedances of $(30 - j\,12)\,\Omega$ are formed into a delta connection and connected to the supply. Calculate the phase current and line currents that flow. Provide magnitudes of both currents and angle of a phase current relative to a phase voltage at 0°. [4]

[Standard calculation aimed at testing basic ability to work with three-phase systems]

Line and phase voltages are equivalent in delta.

$$I_P = \frac{V_P}{Z} = \frac{400}{30 - j20} = 12.38\angle 21.8°\ \text{A}$$

[1 mark for method; 1 mark for magnitude and 1 mark for angle including correct (positive) sign.]

$$|I_L| = \sqrt{3}\,|I_P| = 21.44\,A$$

ii) Calculate the real and reactive powers drawn by the delta-connected impedances. [4]

[Standard calculation aimed at testing basic ability to work with real and reactive power]

Take "power factor" angle as the angle of the impedance from part (i), that is -21.8°, or use negation of angle of current.

Answers

$$P = \sqrt{3}\, V_L I_L cos(\theta) = \sqrt{3} \times 400 \times 21.44\, cos(-21.8°) = 13.79\, kW$$
$$Q = \sqrt{3}\, V_L I_L sin(\theta) = \sqrt{3} \times 400 \times 21.44\, sin(-21.8°) = -5.52\, kVAr$$

[1 mark for formulas; 2 marks for accurate answers; 1 mark for correct units for both answers.]

d) Consider a typical photovoltaic (PV) panel.

i) Explain why not all photons of light that impinge on the panel are captured and converted to usable electrical energy. [4]

[Interpretation of notes to answer question.]

[2 marks for discussion of band-gap energy issues and 2 marks for other factors].

Features of panel construction prevent some photons reaching the active part of the panel, the junction. These include reflections at the protective glass panel or surface of the silicon and shadowing by the top-side metallisation strips or absorption by transparent conductive contact.

Photons that reach the junction only create a hole-electron pair if the photon energy exceeds the band-gap energy thus lower energy (longer wavelength) photons are not captured and converted. Higher energy photons lose energy in excess of band-gap to heat so only the band-gap energy is captured as useful electrical energy.

ii) It is proposed to increase the efficiency of a PV panel by creating two *p-n* junctions, one in front of the other, with different band-gaps. Explain why this might be thought to increase efficiency **and** explain whether the low or high band-gap junction should be placed first in the path of the incident light. [4]

[Interpretation of notes to answer question which make only brief mention of multi-junction cells.]

The band-gaps of the two junctions can be chosen to capture photons of different energy, one of which is set to capture efficiently light at the red end of the spectrum, a low band-gap energy, and one set to capture light effectively at the blue end of the spectrum, a high band-gap energy. Being individually tailored to the two parts of the spectrum, fewer photons fail to be captured and fewer photons have energy well in excess of the band-gap which is wasted.

[2 marks].

If the low energy gap is placed first it captures low and high energy photons and wastes much of the energy of the high energy photons. It is better to place the high energy gap first and capture high-energy photons efficiently; low-energy photons will pass through to be captured by the second junction.

[2 marks].

iii) A particular photovoltaic panel produces a short-circuit current of 2.5 A under an irradiance of 1,000 W/m^2. The reverse saturation current of the panel was measured as 2×10^{-10} A at 30°C. Calculate the open circuit voltage under the same irradiance and at a temperature of 30°C. You may assume that the shunt resistance of the panel is negligible and that the ideality factor is 1. [4]

[Use standard model equations.]

[2 marks for inverting standard equation correctly; 2 marks for accurate answer]

$$I_{AK} = I_0 \left[e^{\frac{q\, V_{AK}}{K_I\, k\, T}} - 1 \right]$$

Answers

Rearranging the exponential diode equation will provide the diode voltage.

$$V_{AK} = K_I \frac{kT}{q} ln\left[\frac{I_{AK}}{I_0} + 1\right]$$

Use $q = 1.6\times10^{-19}$ C and $k = 1.38\times10^{-23}$ J/K from formula sheet

$T = 273 + 30 = 303$ K

$$V_{AK} = 1 \times \frac{1.38 \times 10^{-23} \times 303}{1.6 \times 10^{-19}} ln\left[\frac{2.50}{2 \times 10^{-10}} + 1\right] = 0.06075\ V$$

[Note on suitability for on-line assessment. Past exams have used this multi-part question to do some testing of recall of basic facts but have mostly asked for explanations that require some interpretation of facts or some straightforward calculations. Here, the recall element has been avoided.]

Answers

2.

A **boost** switch-mode power supply (SMPS) is to be designed for an input voltage of 10 V and output voltage of 30 V. The SMPS will be required to process a maximum throughput power of 40 W and have an output voltage ripple which is no greater than 75 mV.

The SMPS will be designed to operate in continuous conduction mode down to a throughput power of 0.5 W.

The on-state resistance of the chosen MOSFET is 0.05 Ω. The turn-on and turn-off energy losses of the MOSFET are 2 µJ and 3 µJ when operated at 1 A and scale linearly with current.

i) Calculate the value of the input current when the circuit operates at maximum power. [2]

[Apply standard equation.]

[1 mark for equation and 1 for accurate answer.]

$$I_I = \frac{P_{Max}}{V_I} = \frac{40}{10} = 4.00\ A$$

ii) Calculate the value of duty-cycle at which the SMPS should be operated under continuous conduction. [2]

[Apply standard formula.]

[2 marks for accurate answer.]

$$\delta = \frac{V_O - V_I}{V_O} = \frac{30 - 10}{30} = 0.67$$

iii) Sketch graphs of the current through the inductor, the diode and the capacitor for operation at maximum power. [4]

[Interpretation since waveforms given in notes for Buck case only.]

[1 mark for inductor current which is DC with triangular ripple superimposed; 1 mark for diode which is trapezoid pulses with a downward slope; 2 marks for capacitor current of the same shape as diode but offset negatively by output current and average of zero.]

iv) Calculate the switching frequency that should be chosen to keep the power dissipation in the MOSFET to a maximum of 1 W. [4]

[Standard method but with current dependency of switching losses added.]

[1 mark for formula, 1 mark for scaling loss; 2 marks for accurate answer.]

Maximum dissipation in the MOSFET will be at maximum throughput power for the circuit.

Scale switching energy loss.

Turn on loss = 2 µJ × 1.33/1.00 = 2.67 µJ

Turn off loss = 3 µJ × 1.33/1.00 = 4.00 µJ

$$f = \frac{P_{Loss}^{Max} - \delta I_{DS}^2 R_{DS}}{E_{On} + E_{Off}} = \frac{1 - 0.67 \times 4^2 \times 0.05}{2.67 \times 10^{-6} + 4 \times 10^{-6}} = 69.96\ kHz$$

Answers

v) It is required that the circuit operate in continuous mode down to a throughput power of 0.5 W. Calculate the value of inductor that should be chosen to achieve this. [6]

[Two-step method with both steps common on their own.]

Find critical **input** current [2 marks.]

$$I_i^{Crit} = \frac{P^{Crit}}{V_I} = \frac{0.5}{10} = 0.05\ A$$

Ripple current can be no more than twice this [2 marks].

$$\Delta i_L \leq 2\ I_i^{Crit} \leq 0.1\ A$$

Choose inductor for this ripple value.

$$L = \frac{V_I\ \delta}{\Delta i_L\ f} = \frac{10 \times 0.67}{0.1 \times 69.96 \times 10^3} = 958\ \mu H$$

vi) It is intended to use an output-side capacitor of 470 µF. Calculate the maximum value of effective series resistance (ESR) that this capacitor could be allowed to have. You may approximate where needed. [8]

[Variation of a standard method in which the steps need to be applied in a different order.]

Maximum ESR will be set by total ripple voltage minus that arising across the capacitor.

Worst case is at largest output current which is $I_O^{Max} = \frac{P^{Max}}{V_O} = \frac{40}{30} = 1.33\ A$ [1 mark]

$$\Delta v_C = \frac{I_O(1-\delta)}{f\ C} = \frac{1.33 \times (1-0.67)}{69.96 \times 10^3 \times 470 \times 10^{-6}} = 13\ mV$$

[2 marks for v_C equation; 2 for accurate answer]

Allowed ripple across ESR is therefore (approximately, ignoring waveform shape and phase angle issues) 75 mV – 13 mV = 62 mV [1 mark]

$$R_{ESR} = \frac{\Delta v_{ESR}}{I_I + ½\Delta i_L} = \frac{0.062}{4 + ½ \times 0.1} = 15.3\ m\Omega$$

[1 mark for equation; 1 for accurate answer]

vii) This SMPS will be used with a feedback control system that will set duty-cycle to achieve the desired output voltage. Consider the case of a throughput of 0.25 W. Calculate the duty-cycle the controller would need to apply to maintain the output voltage at 30 V. [4]

[This problem is unfamiliar since no similar questions were set in class.]

First, recognise that the circuit will be in discontinuous operation and that is why the duty-cycle will differ from that in (ii). The equation that applies (chosen from formula sheet) is:

$$\frac{V_O}{V_I} = \frac{1}{1 - \frac{V_I \delta^2}{2fL\ I_I}}$$

Re-arrange to find duty-cycle.

$$V_O - V_O \frac{V_I \delta^2}{2fL\ I_I} = V_I$$

Answers

$$\delta^2 = (V_O - V_I)\frac{2fL\,I_I}{V_O V_I}$$

[2 mark for selecting equation and re-arrangement.]

Need to find input current: $I_I = \frac{P}{V_I} = \frac{0.25}{10} = 0.025\ A$ [1 mark]

$$\delta^2 = (V_O - V_I)\frac{2fL\,I_I}{V_O V_I} = (30-10)\frac{2 \times 69.96 \times 10^3 \times 958 \times 10^{-6} \times 0.025}{30 \times 10} = 0.223$$

$$\delta = 0.473$$

[1 mark for accurate answer.]

[Note on suitability for on-line assessment. There are various subtle differences between Buck and Boost SMPS and several ways to set up the question such as analysis of given circuit or design to specification; specified operating frequency or find frequency from thermal conditions. Ill-prepared students would waste time trying to find a question that closely resembles this one. This question has some standard sections which most students would get right and where ill-prepared students would benefit from sight of the notes to confirm equations. However, there are two unusual features: a question about voltage ripple pitched differently to the notes and a numerical question on discontinuous mode. Neither has been seen like this in the class examples or the past papers. Students who are on top of the material should be able to see what to do and make a good attempt at an answer.]

Answers

3. An induction machine which is a 3-phase, 2 pole-pair, star-connected machine is connected to a 50 Hz supply with a phase-voltage of 230 V. The parameters of the equivalent circuit, referred to the stator, are:

Stator resistance,	R_S = 1.4 Ω.
Stator leakage reactance,	X_S = 2.0 Ω;
Magnetising resistance,	R_M = 200 Ω;
Magnetising reactance,	X_M = 40 Ω;
Referred rotor leakage reactance,	X_R = 1.5 Ω;
Referred rotor resistance,	R_R = 0.7 Ω;

The machine is operated with a slip of s = 0.02.

i) Give an expression for the total input impedance of the machine and calculate its value. [8]

[4 marks for correct formulation of impedance, including correct parallel terms and inclusion of s]

Total impedance: $Z_T = R_S + jX_S + \left(R_M // jX_M // \left(\frac{R'_R}{s} + jX'_R\right)\right)$

[4 marks for accurate computation.]

$$Z_T = 19.91 + j16.48 = 25.84\angle 39.62° \ \Omega$$

ii) Calculate the stator current and referred rotor current. [5]

[Answer from previous part will need to be carried forward; full marks can be obtained here if method is sound.]

First, find stator current from the phase voltage $I_S = \frac{V_S}{Z_T} = \frac{230}{25.84\angle 39.62°} = 20.58 \angle -39.62° \text{ A}$

[2 marks for accurate answer.]

Then rotor current is found from the current divider rule (or otherwise)

$$I_R = I_S \times \left(\frac{R_M // jX_M}{R_M // jX_M + \left(\frac{R'_R}{s} + jX'_R\right)}\right)$$

[1 mark for correct formulation]

I_R = 5.97 ∠-4.03° A.

[2 mark for accurate answer.]

iii) Calculate the electro-magnetic torque developed. [5]

[Answer from previous part will need to be carried forward; full marks can be obtained here if method is sound.]

Power from standard formula $P_{EM} = 3\, I'^2_R\, R'_R \left(\frac{1-s}{s}\right) = 3 \times 5.97 \times 0.7 \times \left(\frac{1-0.02}{0.02}\right) = 3.67 \text{ kW}$

[1 mark for correct formula; 1 mark for accurate answer]

Answers

Torque found by dividing by rotor speed, which is $\omega_R = \frac{2\pi f}{P}(1-s) = 153.94\, rad/s$

$$T = \frac{P_{EM}}{\omega_R} = \frac{3{,}670}{153.94} = 23.82 \text{ Nm}$$

[1 mark for correct formula; 2 mark for accurate answer]

iv) Re-calculate the electro-magnetic torque developed for the case of s = 0.04 [5]

New torque found by repeating method above. [5 marks for accurate answer.]

$$T = 43.41 \text{ Nm}$$

v) Explain, with reference to the induced voltage in the rotor windings, why the torque is expected to be an approximately linear function of slip for small slip (operation close to synchronous speed) **and** assess the linearity by comparing the answers in parts (iii) and (iv). [7]

[Interpretation of notes]

[5 marks for explanation and 2 mark for comparison]

As slip increases, the rate-of-change of flux-linkage increases linearly and therefore so too does the voltage induced in the rotor.

The rotor voltage will be at a higher frequency at higher slip and so the inductive impedance is larger, but the impedance is dominate by the resistance. The current is therefore a little less than a linear function of slip. The torque is further reduced below a linear increase by an angle shift between rotor current and rotor flux. Thus, we expect an almost but not quite linear increase in torque with slip.

If linear, the torque found in (iv) at 4% slip would be twice that found in (iii) for 2% slip., that is, 47.62 Nm. The calculated torque is 9% below a linear projection from 2% slip.

[Note on suitability for on-line assessment. Well-prepared students who have practiced questions on induction machines normally do well on such an exam question. Those who have not practiced make mistakes in formulating the impedance (omitting s; confusions over parallel R_M and X_M) and in details of calculating rotor current and torque. I would expect these mistakes to persist for students who are ill-prepared even if they have notes available and so the question will still generate a spread of marks. The final part is unusual and needs some fresh thinking from candidates.]

Answers